

Teaching Note:

Cluster-Robust Inference via the Bayesian Bootstrap

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Abstract

How to implement the Bayesian bootstrap for cluster-robust inference? This note develops a cluster-level Dirichlet reweighting framework that delivers a posterior over regression coefficients and variance components. The recommended implementation is the pseudo-score Bayesian bootstrap (PS-BB), which reweights leave-one-cluster-out pseudo-scores rather than raw scores. In simulations, PS-BB tracks the performance of the wild cluster bootstrap closely in small- G designs while preserving the main advantages of the Bayesian bootstrap framework.

1 Introduction

Cluster-robust inference is standard because empirical work often has many observations but only a modest number of independent clusters. The usual frequentist toolbox is familiar: CV1 for baseline cluster-robust inference, CV3 or wild cluster bootstrap when the number of clusters is small, and ad hoc delta-method calculations for nonlinear quantities. Those tools work, but they do not naturally deliver a joint posterior over regression coefficients and derived objects, and they make it awkward to impose simple shape restrictions such as nonnegative variance components.

This paper develops a cluster-level Bayesian bootstrap view of the problem. The core move is simple: treat clusters as the independent observational units, draw Dirichlet weights on clusters, and propagate those weights through the estimator of interest. That perspective

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is useful because it produces an object that looks like a posterior distribution, carries over immediately to nonlinear functionals, and accommodates structured within-cluster covariance models in a transparent way.

The key practical point is that the most literal implementation is not the one we recommend. The *basic* cluster-level Bayesian bootstrap re-solves weighted least squares for each Dirichlet draw. That version is asymptotically correct and clarifies the connection to the usual CRVE, but in small- G settings it inherits the raw-score representation that underlies CV1 and can under-cover materially. Our recommended implementation is the *pseudo-score Bayesian bootstrap* (PS-BB): we compute leave-one-cluster-out pseudo-scores once, then apply Dirichlet reweighting to those pseudo-scores. In simulations, this moves the procedure much closer to CV3 while preserving the main Bayesian-bootstrap advantages.

The second contribution is an extension for settings with credible within-cluster structure. When units inside a cluster can be organized into a hierarchy—for example, neighborhoods within cities within states—we impose exchangeability restrictions on residual cross-products and estimate common covariance components by pooling information across clusters. Dirichlet reweighting then yields a posterior for both β and the variance components. By restricting attention to accepted draws for which the implied covariance components are positive semidefinite, the reported posterior automatically respects the admissible parameter space. This is the cleanest version of the “no negative variances” selling point: the output that is reported to the reader is restricted to feasible covariance decompositions.

The paper is organized around the recommended procedure. Section 2 introduces the clustered regression model. Section 3 presents the cluster-level Dirichlet reweighting framework, first as a basic benchmark and then as the recommended pseudo-score implementation. Section 4 extends the method to variance components and nonlinear functionals. Section 5 summarizes the simulation evidence. Section 6 concludes.

2 Setup

Consider the linear regression model

$$y_g = X_g\beta + u_g, \quad g = 1, \dots, G, \quad (1)$$

where g indexes clusters, X_g is an $N_g \times k$ matrix of regressors, y_g is an N_g -vector of outcomes, and u_g is an N_g -vector of disturbances. Let $N = \sum_{g=1}^G N_g$, stack observations in the usual way, and write the OLS estimator as $\hat{\beta} = (X'X)^{-1}X'y$.

Assumption 1 (Independent clusters). *The cluster-level observations $W_g = (y_g, X_g)$ are independent across $g = 1, \dots, G$.*

Assumption 2 (Cluster-level regularity). *As $G \rightarrow \infty$, the following hold:*

- (i) $G^{-1} \sum_{g=1}^G X_g' X_g \rightarrow Q$ for a positive definite matrix Q .
- (ii) If $s_g = X_g' u_g$, then $G^{-1} \sum_{g=1}^G E[s_g s_g'] \rightarrow \Omega$ for a positive definite matrix Ω .
- (iii) A cluster-level law of large numbers and central limit theorem apply to $\{s_g\}_{g=1}^G$; in particular,

$$G^{-1/2} \sum_{g=1}^G s_g \xrightarrow{d} N(0, \Omega).$$

- (iv) $\max_{1 \leq g \leq G} \|X_g' X_g\|/G \rightarrow 0$ and $E\|s_g\|^{2+\delta} < \infty$ uniformly in g for some $\delta > 0$.

Under Assumptions 1–2,

$$\sqrt{G}(\hat{\beta} - \beta) = \left(G^{-1} \sum_{g=1}^G X_g' X_g \right)^{-1} G^{-1/2} \sum_{g=1}^G s_g \xrightarrow{d} N(0, Q^{-1} \Omega Q^{-1}).$$

The usual cluster-robust estimator is

$$\widehat{\text{Var}}_{\text{CV1}}(\hat{\beta}) = c_G (X' X)^{-1} \left(\sum_{g=1}^G \hat{s}_g \hat{s}_g' \right) (X' X)^{-1}, \quad (2)$$

where $\hat{s}_g = X_g' \hat{u}_g$, $\hat{u}_g = y_g - X_g \hat{\beta}$, and $c_G = G(N-1)/\{(G-1)(N-k)\}$.

3 Cluster-Level Dirichlet Reweighting

3.1 Basic Bayesian bootstrap: useful benchmark, not the default

The Bayesian bootstrap (Rubin, 1981) places a Dirichlet prior on the weights attached to the observed support points. With clustered data, the support points are clusters. The most literal implementation is therefore:

Basic cluster-level BB. For $m = 1, \dots, M$,

1. Draw $w^{(m)} = (w_1^{(m)}, \dots, w_G^{(m)}) \sim \text{Dir}(\alpha, \dots, \alpha)$ and define $\omega_g^{(m)} = G w_g^{(m)}$.

2. Compute the weighted least-squares draw

$$\hat{\beta}_{\text{BB}}^{(m)} = \left(\sum_{g=1}^G \omega_g^{(m)} X_g' X_g \right)^{-1} \left(\sum_{g=1}^G \omega_g^{(m)} X_g' y_g \right). \quad (3)$$

The empirical distribution of $\{\hat{\beta}_{\text{BB}}^{(m)}\}_{m=1}^M$ is the basic BB posterior for β .

The appeal of this construction is conceptual clarity. It is exactly the cluster-level analog of Rubin's Bayesian bootstrap, and it has a clean first-order connection to the usual cluster sandwich.

Proposition 1 (First-order posterior variance of the basic BB). *Let $\hat{s}_g = X_g' \hat{u}_g$ and suppose $\omega_g = G w_g$ with $(w_1, \dots, w_G) \sim \text{Dir}(\alpha, \dots, \alpha)$. Then, to first order,*

$$\text{Var}_w(\hat{\beta}_{\text{BB}}^{(m)} \mid \text{data}) = \frac{G}{G\alpha + 1} (X'X)^{-1} \left(\sum_{g=1}^G \hat{s}_g \hat{s}_g' \right) (X'X)^{-1}. \quad (4)$$

In particular, with the flat prior $\alpha = 1$, the factor is $G/(G + 1)$.

Proof. A first-order expansion of (3) around $\omega_g = 1$ gives

$$\hat{\beta}_{\text{BB}}^{(m)} - \hat{\beta} \approx (X'X)^{-1} \sum_{g=1}^G (\omega_g^{(m)} - 1) \hat{s}_g,$$

using $\sum_g \hat{s}_g = 0$. For symmetric Dirichlet weights,

$$\text{Var}(\omega_g) = \frac{G-1}{G\alpha+1}, \quad \text{Cov}(\omega_g, \omega_h) = -\frac{1}{G\alpha+1} \quad (g \neq h).$$

Therefore,

$$\begin{aligned} \text{Var}_w \left(\sum_g (\omega_g - 1) \hat{s}_g \right) &= \frac{G-1}{G\alpha+1} \sum_g \hat{s}_g \hat{s}_g' - \frac{1}{G\alpha+1} \sum_{g \neq h} \hat{s}_g \hat{s}_h' \\ &= \frac{G-1}{G\alpha+1} \sum_g \hat{s}_g \hat{s}_g' + \frac{1}{G\alpha+1} \sum_g \hat{s}_g \hat{s}_g' \\ &= \frac{G}{G\alpha+1} \sum_g \hat{s}_g \hat{s}_g', \end{aligned}$$

where the second line uses $\sum_g \hat{s}_g = 0$. □

Two implications matter. First, the basic BB is asymptotically aligned with the cluster sandwich. Second, with $\alpha = 1$ its first-order variance is scaled by $G/(G + 1)$ rather than

roughly $G/(G-1)$, so in small- G settings it is typically *less* dispersed than CV1, not more. That is exactly why it is useful to keep the basic BB in the paper as a benchmark but not as the recommended procedure.

A convenient calibration is to choose

$$\alpha^* = \frac{G-2}{G}, \quad (5)$$

which makes $G/(G\alpha^* + 1) = G/(G-1)$ and therefore aligns the Dirichlet scaling with the leading CV1 factor. This corrects a common temptation to appeal to the Jeffreys value $\alpha = 1/2$: for general G , $\alpha = 1/2$ yields $2G/(G+2)$, not $G/(G-1)$.

3.2 Why the recommended procedure uses pseudo-scores

The basic BB reweights *raw* cluster score contributions. In small samples, that is not ideal because the raw OLS residuals inherit leverage distortion from the projection matrix. High-leverage clusters are shrunk too aggressively, and that distortion feeds directly into the meat of the sandwich. The logic is the same as the gap between CV1 and CV3.

This suggests a better finite-sample implementation. Let $\hat{\beta}^{(-g)}$ be the leave-one-cluster-out OLS estimator,

$$\hat{\beta}^{(-g)} = (X'X - X'_gX_g)^{-1}(X'y - X'_gy_g), \quad (6)$$

and define the pseudo-score

$$\check{s}_g = X'X(\hat{\beta} - \hat{\beta}^{(-g)}). \quad (7)$$

These pseudo-scores are the leave-one-cluster-out contributions that underlie jackknife-style cluster-robust inference. In the few-cluster case they restore the influence of high-leverage clusters that raw residual scores understate. For completeness, the corresponding CV3 variance estimator can be written in pseudo-score form as

$$\widehat{\text{Var}}_{\text{CV3}}(\hat{\beta}) = \frac{G-1}{G}(X'X)^{-1} \left(\sum_{g=1}^G \check{s}_g \check{s}'_g \right) (X'X)^{-1}. \quad (8)$$

Equation (8) makes the connection to PS-BB transparent: PS-BB keeps the same leave-one-cluster-out influence representation and replaces the fixed quadratic form with cluster-level Dirichlet reweighting.

3.3 Recommended procedure: the pseudo-score Bayesian bootstrap

Our default recommendation for coefficient inference is therefore the following.

Pseudo-score Bayesian bootstrap (PS-BB).

1. Compute $\hat{\beta}$ and the leave-one-cluster-out estimators $\{\hat{\beta}^{(-g)}\}_{g=1}^G$.
2. Form pseudo-scores $\check{s}_g = X'X(\hat{\beta} - \hat{\beta}^{(-g)})$.
3. For $m = 1, \dots, M$, draw $w^{(m)} \sim \text{Dir}(\alpha^*, \dots, \alpha^*)$, set $\omega_g^{(m)} = Gw_g^{(m)}$, and compute the one-step draw

$$\hat{\beta}_{\text{PS}}^{(m)} = \hat{\beta} + (X'X)^{-1} \sum_{g=1}^G (\omega_g^{(m)} - 1) \check{s}_g. \quad (9)$$

4. For scalar parameters, report either the equal-tailed posterior interval from the draws or the t -calibrated interval

$$\hat{\beta}_j \pm t_{G-1, 1-\tau/2} \widehat{\text{sd}}(\hat{\beta}_{\text{PS},j}^{(1)}, \dots, \hat{\beta}_{\text{PS},j}^{(M)}). \quad (10)$$

The attraction of PS-BB is practical rather than mystical. It keeps the Dirichlet resampling at the cluster level, but it reweights a better influence representation. That is why it tracks CV3 much more closely in the few-cluster designs that matter empirically.

Three remarks are worth making.

Remark 1 (Why this is still a Bayesian bootstrap). *All posterior uncertainty still comes from Dirichlet weights on clusters. What changes is not the resampling mechanism but the object that is being reweighted: the basic BB reweights raw weighted-least-squares contributions, while PS-BB reweights jackknife-corrected cluster influence contributions.*

Remark 2 (Computation). *Once the leave-one-cluster-out estimators are computed, each PS-BB draw is cheap: no matrix inversion is required inside the simulation loop. This makes PS-BB substantially lighter than re-solving weighted least squares for every draw.*

Remark 3 (Why we do not studentize draw by draw). *In the basic BB, Dirichlet weights change the effective design matrix itself, so a per-draw standard error estimates the variance of a differently weighted estimator rather than the variance of the original estimator under a perturbed score realization. That is why draw-by-draw studentization is not the right correction here. The pseudo-score formulation avoids this problem by keeping the design fixed and putting the small-sample adjustment in the influence representation and the $t(G-1)$ critical value.*

3.4 First-order validity

The formal first-order justification is straightforward: cluster-level Dirichlet reweighting reproduces the same asymptotic distribution as the cluster-level sampling problem.

Proposition 2 (Asymptotic validity of cluster-level Dirichlet reweighting). *Under Assumptions 1–2, the basic BB posterior for $\sqrt{G}(\hat{\beta}_{\text{BB}}^{(m)} - \hat{\beta})$ converges in distribution, in probability over the data, to $N(0, Q^{-1}\Omega Q^{-1})$. Consequently, equal-tailed credible intervals from the basic BB are asymptotically valid frequentist confidence intervals.*

Proof sketch. The weighted estimating equation

$$\sum_{g=1}^G \omega_g X_g' (y_g - X_g \beta) = 0$$

induces the same first-order linearization as the unweighted cluster estimating equation, with the weighted empirical process replacing the unweighted one. Under the independent-cluster law of large numbers and central limit theorem in Assumption 2, the Dirichlet-weighted empirical process has the same Gaussian limit as the sampling process. This is the cluster-level version of the Bayesian-bootstrap result for estimating equations in Chamberlain and Imbens (2003). The statement here is phrased for independent clusters and a cluster-level CLT; no identical-distribution assumption is needed beyond those high-level conditions. $\square$

For PS-BB, the point is more modest and more practical: pseudo-scores are an asymptotically equivalent influence representation, but they are chosen because they behave better in the few-cluster case. The paper’s recommendation is therefore empirical and operational: use PS-BB as the default, and keep the basic BB in the paper as the conceptual benchmark.

4 Structured Within-Cluster Covariance

4.1 Exchangeability decomposition

Now suppose each cluster i contains subgroups $j = 1, \dots, J_i$ and units $k = 1, \dots, N_{ij}$. We allow $d \geq 1$ outcomes per unit and write the residual vector as $U_{ijk} = Y_{ijk} - X'_{ijk}\beta \in \mathbb{R}^d$. We impose three exchangeability restrictions on second moments:

$$E[U_{ijk}U'_{ijk}] = A, \tag{11}$$

$$E[U_{ijk}U'_{ijl}] = B \quad (k \neq l), \tag{12}$$

$$E[U_{ijk}U'_{ilm}] = C \quad (j \neq l). \tag{13}$$

When these matrices satisfy

$$\Sigma_S = C, \quad \Sigma_V = B - C, \quad \Sigma_\varepsilon = A - B, \quad (14)$$

with each component positive semidefinite, the restrictions are equivalent to the random-effects decomposition

$$U_{ijk} = S_i + V_{ij} + \varepsilon_{ijk}, \quad (15)$$

where S_i , V_{ij} , and ε_{ijk} are mutually uncorrelated mean-zero d -vectors with covariance matrices Σ_S , Σ_V , and Σ_ε .

This is the part of the framework that creates new estimands. The basic coefficient posterior only needs cluster independence. The exchangeability restrictions are optional structure that let us estimate and report meaningful covariance components.

4.2 Extended PS-BB algorithm

To keep the weighting internally coherent, we estimate A , B , and C by pooled cross-products at each draw rather than by mixing cluster-level averaging formulas with pair-level pooling. Define the cluster-specific counts

$$Q_{1i} = \sum_{j=1}^{J_i} N_{ij}, \quad Q_{2i} = \sum_{j=1}^{J_i} N_{ij}(N_{ij} - 1), \quad Q_{3i} = \sum_{j \neq l} N_{ij}N_{il}.$$

For each accepted coefficient draw, compute residuals and then the pooled estimators

$$A^{(m)} = \frac{\sum_{i=1}^G \omega_i^{(m)} \sum_{j=1}^{J_i} \sum_{k=1}^{N_{ij}} \hat{U}_{ijk}^{(m)} \hat{U}_{ijk}^{(m)'}}{\sum_{i=1}^G \omega_i^{(m)} Q_{1i}}, \quad (16)$$

$$B^{(m)} = \frac{\sum_{i=1}^G \omega_i^{(m)} \sum_{j=1}^{J_i} \sum_{\substack{k,l=1 \\ k \neq l}}^{N_{ij}} \hat{U}_{ijk}^{(m)} \hat{U}_{ijl}^{(m)'}}{\sum_{i=1}^G \omega_i^{(m)} Q_{2i}}, \quad (17)$$

$$C^{(m)} = \frac{\sum_{i=1}^G \omega_i^{(m)} \sum_{\substack{j,l=1 \\ j \neq l}}^{J_i} \sum_{k=1}^{N_{ij}} \sum_{m'=1}^{N_{il}} \hat{U}_{ijk}^{(m)} \hat{U}_{ilm'}^{(m)'}}{\sum_{i=1}^G \omega_i^{(m)} Q_{3i}}. \quad (18)$$

Then form

$$\Sigma_S^{(m)} = C^{(m)}, \quad \Sigma_V^{(m)} = B^{(m)} - C^{(m)}, \quad \Sigma_\varepsilon^{(m)} = A^{(m)} - B^{(m)}. \quad (19)$$

This algorithm gives a posterior draw for the full parameter vector $(\beta, \Sigma_S, \Sigma_V, \Sigma_\varepsilon)$. In

practice we recommend using PS-BB draws for β in the first stage and then (16)–(18) in the second stage.

4.3 Admissibility and the “no negative variances” point

The most useful way to present the covariance-component extension is not to claim that the raw draw formulas can never wander outside the feasible set. They can. The right claim is stronger and cleaner: the *reported posterior* is restricted to the admissible covariance cone.

Concretely, after computing (19), keep draw m if and only if the matrices $\Sigma_S^{(m)}$, $\Sigma_V^{(m)}$, and $\Sigma_\varepsilon^{(m)}$ are all positive semidefinite. For scalar outcomes, this reduces to checking three nonnegative numbers. For multivariate outcomes, a numerical implementation should allow a small tolerance when checking eigenvalues.

This accept/reject step does two jobs.

- (i) It ensures that the posterior over variance components is supported only on feasible covariance decompositions. That is the precise sense in which the method avoids reporting negative variances.
- (ii) It acts as a diagnostic. A low acceptance rate means either that the data are pushing some component toward the boundary or that the exchangeability restrictions are straining against the data.

4.4 Nonlinear functionals and optional GLS refinement

Once accepted draws are available, nonlinear functionals are immediate. If $d = 2$ and Σ_V is the city-level covariance matrix, then the effect on outcome 2 of a one-standard-deviation increase in the city effect on outcome 1 is simply $\sigma_{V,12}/\sqrt{\sigma_{V,11}}$ draw by draw. No delta method is needed.

The variance-component estimates can also be used to build a working covariance matrix $\hat{\Omega}_i$ and replace OLS by GLS inside the coefficient step:

$$\hat{\beta}_{\text{GLS}}^{(m)} = \left(\sum_{i=1}^G \omega_i^{(m)} X_i' \hat{\Omega}_i^{-1} X_i \right)^{-1} \left(\sum_{i=1}^G \omega_i^{(m)} X_i' \hat{\Omega}_i^{-1} y_i \right). \quad (20)$$

This is an efficiency refinement, not a robustness requirement. Robustness still comes from the cluster-level Dirichlet reweighting; GLS simply uses the within-cluster structure more aggressively when the structure is credible.

5 Simulation Evidence

This section keeps the simulation evidence focused on the recommendation. The basic BB is shown as a benchmark. The headline comparison is between CV1, CV3, BB-OLS, and the two PS-BB intervals.

5.1 Design

The design is the same as in the original draft. Each Monte Carlo dataset has a three-level hierarchy (state $\supset$ city $\supset$ individual), a treatment assigned at the state level, and either a scalar or a bivariate outcome. All experiments use 500 simulation replications and $M = 999$ posterior draws per replication.

For the scalar-outcome designs, the data-generating process is

$$Y_{ijk} = 1 + 0.5D_i + 0.3Z_{ijk} + S_i + V_{ij} + \varepsilon_{ijk},$$

with half the states treated and with three configurations for the variance components:

Configuration	σ_S	σ_V	σ_ε
A (Balanced)	1.0	1.0	1.0
B (State-dominant)	2.0	0.5	1.0
C (City-dominant)	0.3	1.5	1.0

For the bivariate design, the true covariance components are

$$\Sigma_S = \begin{pmatrix} 0.50 & 0.20 \\ 0.20 & 0.30 \end{pmatrix}, \quad \Sigma_V = \begin{pmatrix} 1.00 & 0.40 \\ 0.40 & 0.80 \end{pmatrix}, \quad \Sigma_\varepsilon = \begin{pmatrix} 1.00 & 0.10 \\ 0.10 & 1.00 \end{pmatrix}.$$

5.2 Coefficient inference

Table 1 shows the many-cluster case and Table 2 the few-cluster case.

Table 1: Coverage (%) and average width of 95% intervals for β_1 , $G = 50$ clusters.

Configuration	CV1		CV3		BB-OLS		PS-BB(pct)		PS-BB(t)	
	Cov	Width	Cov	Width	Cov	Width	Cov	Width	Cov	Width
A (Balanced)	94.2	1.239	95.6	1.275	92.6	1.162	94.6	1.268	95.6	1.300
B (State-dom.)	95.0	2.361	95.6	2.431	93.4	2.215	94.8	2.422	95.4	2.480
C (City-dom.)	93.6	0.674	94.6	0.693	92.8	0.634	94.4	0.686	95.0	0.705

Table 2: Coverage (%) and average width of 95% intervals for β_1 , $G = 15$ clusters.

Configuration	CV1		CV3		BB-OLS		PS-BB(pct)		PS-BB(t)	
	Cov	Width	Cov	Width	Cov	Width	Cov	Width	Cov	Width
A (Balanced)	92.6	2.346	94.6	2.597	88.2	1.920	93.6	2.549	95.4	2.779
B (State-dom.)	91.0	4.308	92.8	4.775	86.4	3.522	92.4	4.685	95.2	5.110
C (City-dom.)	92.8	1.543	95.2	1.699	88.0	1.271	94.2	1.664	95.6	1.817

The message is simple. With many clusters, the basic BB is usable but still a bit too narrow relative to CV3. With few clusters, the basic BB under-covers materially, while PS-BB nearly matches CV3 and the t -calibrated PS-BB slightly exceeds it. That is exactly the pattern one would hope to see if the pseudo-score representation is fixing the right finite-sample problem.

5.3 Variance components and nonlinear functionals

Table 3 reports the bivariate variance-component results under the extended procedure.

Table 3: Variance-component recovery ($d = 2$, $G = 50$). Acceptance rate: 100.0%.

Parameter	True	Post. Mean	RMSE	95% Cov. (%)
$\sigma_{S,11}$	0.50	0.473	0.125	89.0
$\sigma_{S,12}$	0.20	0.187	0.079	90.6
$\sigma_{S,22}$	0.30	0.283	0.083	88.4
$\sigma_{V,11}$	1.00	0.993	0.093	92.4
$\sigma_{V,12}$	0.40	0.391	0.062	93.6
$\sigma_{V,22}$	0.80	0.790	0.071	92.8
$\sigma_{\varepsilon,11}$	1.00	1.000	0.033	94.4
$\sigma_{\varepsilon,12}$	0.10	0.102	0.023	92.8
$\sigma_{\varepsilon,22}$	1.00	1.003	0.025	94.2

The acceptance rate is 100% in this design, so the admissibility restriction is not binding. The more important takeaway is qualitative: the posterior handles covariance components and nonlinear functionals directly. For example, the functional $\sigma_{V,12}/\sqrt{\sigma_{V,11}}$ is centered close to its true value of 0.40 with empirical coverage near nominal. This is precisely the kind of estimand for which the posterior approach is cleaner than a delta-method add-on.

6 Discussion

The paper reads best when it is organized around one main recommendation:

Use cluster-level Dirichlet reweighting as the framework, use PS-BB as the default implementation for coefficient inference, and use the structured covariance extension only when the within-cluster exchangeability restrictions are substantively credible.

That presentation is stronger than starting with the basic BB as the headline method and then retreating from it later. The basic BB still belongs in the paper because it clarifies the connection to cluster-robust variance estimation and because it is asymptotically correct. But its role is explanatory and benchmark-oriented. The paper’s practical message is the pseudo-score version.

Relative to existing methods, the paper’s comparative advantage is not that it replaces CV3 or the wild cluster bootstrap on every dimension. The advantage is that it gives a single cluster-level resampling framework that delivers coefficient inference, nonlinear functionals, and structured covariance components in one place. That is a useful addition to the applied toolkit.

Several extensions remain natural. Instrumental variables can be handled by replacing OLS score equations with the corresponding IV estimating equations. More flexible within-cluster covariance models can replace the simple exchangeability restrictions. And the calibration of the Dirichlet concentration parameter beyond $\alpha^* = (G - 2)/G$ is still open territory. But the main practical result is already clear: if the goal is a readable, defensible, and practically useful paper, PS-BB should be the method in the foreground.

References

- Chamberlain, G. and G. W. Imbens (2003). Nonparametric applications of Bayesian inference. *Journal of Business & Economic Statistics*, 21(1), 12–18.
- Rubin, D. B. (1981). The Bayesian bootstrap. *Annals of Statistics*, 9(1), 130–134.